Introduction to Proofs

Complete Exam Preparation Guide

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EXAM TOMORROW — MCQ + Written

All 9 proof types covered · Templates included · 20 practice questions at the end

Logic · Direct · Contrapositive · Contradiction · Cases · Existence · Uniqueness · Vacuous

§1 Introduction to Proofs

▶ Definition: Mathematical Proof

A **proof** is a rigorous logical argument that establishes the truth of a mathematical statement beyond all doubt. It proceeds from known axioms, definitions, and previously proved theorems using valid rules of inference.

0.1 Key Terminology

Term	Meaning
Axiom	A statement accepted as true without proof (foundation)
Theorem	An important statement that has been proved
Lemma	A "helper" theorem used to prove a bigger result
Corollary	A result that follows directly from a theorem
Conjecture	A statement believed true but not yet proved
Proposition	$P \Rightarrow Q$ — if hypothesis P, then conclusion Q
Hypothesis (P)	The "given" / "assume" part
Conclusion (Q)	What we must show / prove

0.2 Logical Foundation

▶ Definition: Implication $P \Rightarrow Q$ and its Forms

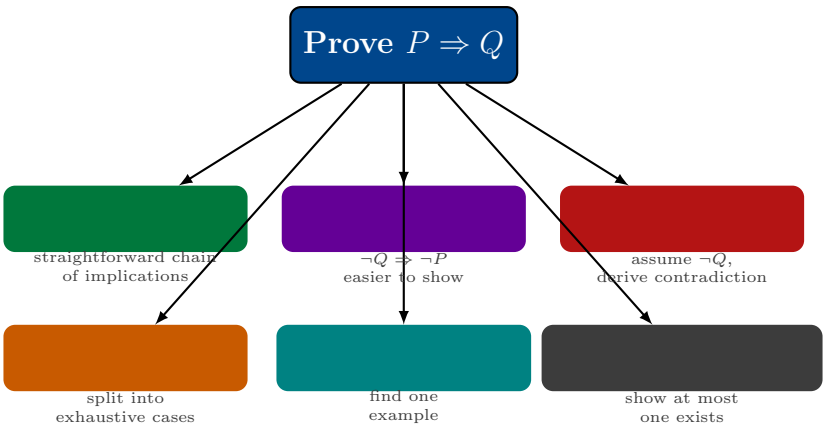
Name	Form	Logically equivalent?
Original	$P \Rightarrow Q$	—
Converse	$Q \Rightarrow P$	**NOT equivalent**
Inverse	$\neg P \Rightarrow \neg Q$	**NOT equivalent**
Contrapositive	$\neg Q \Rightarrow \neg P$	**YES, equivalent**

★ Key Exam Point: Must Memorise

$P \Rightarrow Q \equiv \neg Q \Rightarrow \neg P$ (contrapositive always equivalent)
$P \Rightarrow Q \not\equiv Q \Rightarrow P$ (converse is NOT the same!)

§2 Proof Methods & Strategy

Choosing a Proof Method



Note: Strategy Guide

- **Try Direct first** — if P gives you enough to work with
- **Try Contrapositive** — when $\neg Q$ is a strong, useful hypothesis
- **Try Contradiction** — when you can't see a path forward, or for "irrational/impossible" statements
- **Try Cases** — when the domain splits naturally (even/odd, positive/negative, $=/\neq$)
- **Existence** — when asked "there exists..."
- **Uniqueness** — when asked "there is exactly one..."

§3 Direct Proof

▶ Definition: Direct Proof

To prove $P \Rightarrow Q$ directly:

1. **Assume** P is true
2. Use definitions, algebra, and logic to **derive** Q step by step
3. **Conclude** Q holds

Proof Template

Assume P. [State what P gives you]
Then [step 1] because [reason].
Therefore [step 2] because [reason].
⋮
Hence Q. ■

Example: If n is even, then n^2 is even

Proof. Assume n is even. Then $n = 2k$ for some integer k.
Therefore $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$.
Since $2k^2 \in \mathbb{Z}$, n^2 is even. ■

Example: If $a \mid b$ and $b \mid c$, then $a \mid c$ (transitivity of divisibility)

Proof. Assume $a \mid b$ and $b \mid c$.
Then $b = am$ and $c = bn$ for some integers m, n.
Substituting: $c = (am)n = a(mn)$.
Since $mn \in \mathbb{Z}$, we have $a \mid c$. ■

★ Key Exam Point: Direct Proof — What to Watch

- Always **start by assuming** P
- When "n is even": write $n = 2k$; when "n is odd": write $n = 2k + 1$
- When "a divides b" ($a \mid b$): write $b = ak$ for some $k \in \mathbb{Z}$
- End with a clear statement that Q holds, then ■

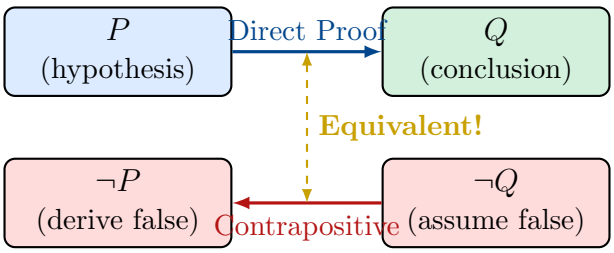
§4 Contrapositive Proof

▶ Definition: Proof by Contrapositive

Instead of proving $P \Rightarrow Q$, prove the logically equivalent statement:

$$\neg Q \Rightarrow \neg P$$

Assume $\neg Q$, then derive $\neg P$.



Proof Template

We prove the contrapositive: $\neg Q \Rightarrow \neg P$.
Assume $\neg Q$. [State what $\neg Q$ gives you]
Then [derive steps]
Therefore $\neg P$ holds. ■

Example: If n^2 is even, then n is even

Proof. We prove the contrapositive: if n is odd, then n^2 is odd.
Assume n is odd. Then $n = 2k + 1$ for some integer k.
$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$.
Since $2k^2 + 2k \in \mathbb{Z}$, n^2 is odd. ■

Example: If $3n + 2$ is odd, then n is odd

Proof. Contrapositive: if n is even, then $3n + 2$ is even.
Assume $n = 2k$. Then $3n + 2 = 6k + 2 = 2(3k + 1)$, which is even. ■

★ Key Exam Point: When to Use Contrapositive

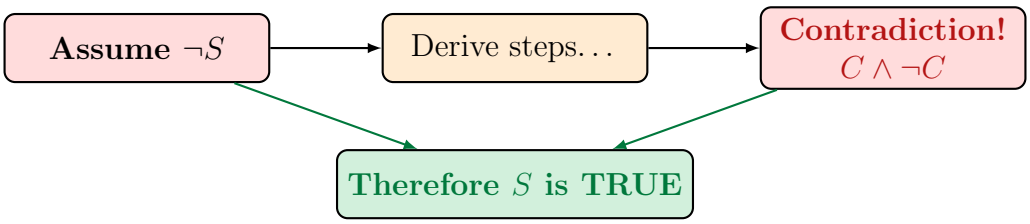
Use contrapositive when the hypothesis $\neg Q$ is **stronger / easier to work with** than P. Classic sign: statement involves "if the result has property X, then the input has property X" (e.g., even/odd, divisibility).

§5 Proof by Contradiction

▶ Definition: Proof by Contradiction (Reductio ad Absurdum)

To prove statement S:

1. **Assume** $\neg S$ (the negation)
2. Derive a **logical contradiction** (e.g., $0 = 1$, p is both even and odd, $r < r$)
3. Conclude that $\neg S$ is false, so S must be true



Proof Template

Suppose for contradiction that $\neg S$.
[Derive consequences of $\neg S$]
But this contradicts [some known fact / earlier step].
Therefore S must be true. ■

Example: $\sqrt{2}$ is irrational

Proof. Suppose for contradiction $\sqrt{2} \in \mathbb{Q}$. Write $\sqrt{2} = p/q$ with $p, q \in \mathbb{Z}$, $q \neq 0$, and $\gcd(p, q) = 1$.

Then $2 = p^2/q^2$, so $p^2 = 2q^2$. Hence p^2 is even, so p is even (contrapositive lemma). Write $p = 2k$. Then $4k^2 = 2q^2 \Rightarrow q^2 = 2k^2$, so q is also even.
But then $\gcd(p, q) \geq 2$, contradicting $\gcd(p, q) = 1$. ■

Example: There are infinitely many primes

Proof. Suppose finitely many primes: $p_1, p_2, \ldots, p_n$.
Let $N = p_1 p_2 \cdots p_n + 1$.
$N > 1$ so N has a prime factor p. But p divides N and $p_1 p_2 \cdots p_n$, so $p \mid 1$, forcing $p = 1$. But 1 is not prime. Contradiction. ■

★ Key Exam Point: Direct vs Contradiction — Key Difference

Direct / Contrapositive	Contradiction
Proves $P \Rightarrow Q$	Can prove any statement S
Assume P (or $\neg Q$) and work forward	Assume $\neg S$, find something impossible
Positive construction	Negative / impossible assumption

§6 Proof by Cases

▶ Definition: Proof by Cases (Exhaustive Proof)

To prove $P \Rightarrow Q$, partition the domain into exhaustive, mutually exclusive cases $C_1, C_2, \ldots, C_n$ and prove Q holds in each case.

$$P \Rightarrow [(C_1 \Rightarrow Q) \wedge (C_2 \Rightarrow Q) \wedge \cdots \wedge (C_n \Rightarrow Q)]$$

Proof Template

Note that [the domain / hypothesis] splits into cases: $C_1, C_2, \ldots, C_n$.

Case 1: C_1. [Prove Q].
Case 2: C_2. [Prove Q].
⋮
Case n: C_n. [Prove Q].

Since all cases give Q, the proof is complete. ■

Example: $n^2 + n$ is even for all $n \in \mathbb{Z}$

Proof. We consider two cases.

Case 1: n is even. Write $n = 2k$. Then $n^2 + n = 4k^2 + 2k = 2(2k^2 + k)$, which is even.

Case 2: n is odd. Write $n = 2k + 1$. Then $n^2 + n = (2k + 1)^2 + (2k + 1) = 4k^2 + 4k + 1 + 2k + 1 = 4k^2 + 6k + 2 = 2(2k^2 + 3k + 1)$, which is even.

In both cases $n^2 + n$ is even. ■

Example: $|xy| = |x||y|$ for all $x, y \in \mathbb{R}$

Proof. Cases: $x \geq 0, y \geq 0$; $x \geq 0, y < 0$; $x < 0, y \geq 0$; $x < 0, y < 0$.
In each case verify $|xy| = |x||y|$ by expanding. (All four cases are symmetric.) ■ ■

★ Key Exam Point: Case Coverage

Cases must be **exhaustive** (cover everything) **and** together cover all possibilities. Common splits: even/odd, positive/zero/negative, $n \bmod 3 \in \{0, 1, 2\}$.

§7 Existence Proof

▶ Definition: Existence Proof

To prove $\exists x\, P(x)$ ("there exists x such that $P(x)$"), either:

- **Constructive:** explicitly *exhibit* an x and verify $P(x)$
- **Non-constructive:** argue existence without finding x (e.g., contradiction, intermediate value)

Proof Template

[**Constructive**] Let $x =$ [specific value]. Then $P(x)$ holds because [verification]. ■

[**Non-constructive**] Suppose no such x exists. [Derive contradiction.] ■

Example: There exist irrational a, b such that a^b is rational

Proof. [Non-constructive] Let $r = \sqrt{2}^{\sqrt{2}}$.
Case 1: r is rational. Take $a = b = \sqrt{2}$; done.
Case 2: r is irrational. Take $a = \sqrt{2}^{\sqrt{2}}$, $b = \sqrt{2}$. Then $a^b = (\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = \sqrt{2}^2 = 2 \in \mathbb{Q}$.
Either way, such a, b exist. ■

Example: Constructive: $\exists$ an integer n such that $n^2 = n$

Proof. Let $n = 1$. Then $n^2 = 1 = n$. ■ ■

Note: Constructive vs Non-constructive

Constructive proofs are generally preferred because they *find* the witness. Non-constructive proofs confirm existence but don't tell you *which* object satisfies the

property.

§8 Uniqueness Proof

▶ Definition: Uniqueness Proof

To prove "there exists a **unique** x such that $P(x)$" (written $\exists! x\, P(x)$):

1. **Existence:** Show at least one such x exists
2. **Uniqueness:** Assume x_1 and x_2 both satisfy P; prove $x_1 = x_2$

Proof Template

Existence: [Exhibit or prove some x_0 satisfies $P(x_0)$.]

Uniqueness: Suppose x_1 and x_2 both satisfy P.
[Derive $x_1 = x_2$ using algebra / logic.]
Therefore such x is unique. ■

Example: The equation $2x + 3 = 7$ has a unique solution

Proof. **Existence:** $x = 2$ satisfies $2(2) + 3 = 7$. ✓

Uniqueness: Suppose $2x_1 + 3 = 7$ and $2x_2 + 3 = 7$.
Subtracting: $2x_1 - 2x_2 = 0 \Rightarrow x_1 = x_2$. ■

Example: Unique additive identity in $\mathbb{R}$

Proof. **Existence:** 0 is an additive identity.
Uniqueness: Suppose e is also an additive identity. Then $e = e + 0 = 0$. ■

★ Key Exam Point: Two-Part Structure

Every uniqueness proof **must** have both parts: existence AND uniqueness. Proving only one is incomplete and will lose marks!

§9 Vacuous Proof

▶ Definition: Vacuous Proof

A proof of $P \Rightarrow Q$ is **vacuously true** when P is **always false**. If P is false, then $P \Rightarrow Q$ is *automatically true* regardless of Q.

$$P \Rightarrow Q \equiv \neg P \vee Q$$

If P is always false, then $\neg P$ is true, making the whole statement true.

Truth Table for $P \Rightarrow Q$

P	Q	$P \Rightarrow Q$	
F	**F**	**T**	Vacuously true (P false)
F	**T**	**T**	
T	**F**	**F**	
T	**T**	**T**	

Example: Vacuous Proof Example

Claim: For all $x \in \mathbb{R}$, if $x^2 < 0$ then $x = 5$.

Proof. The hypothesis "$x^2 < 0$" is never true for real x (squares are non-negative). Since the hypothesis is always false, the implication is vacuously true. ■

Example: Empty Set Example

Claim: Every element of $\emptyset$ is a purple elephant.

Proof. The statement "$x \in \emptyset$" is always false. Hence vacuously true. ■

★ Key Exam Point: Vacuous Truth in Exams

A vacuous proof is **not a trick** — it is logically valid. Recognise it when you see: "for all x, if [impossible condition], then [anything]." The key word is **impossible hypothesis**.

§ Master Summary Table

Method	How It Works	Use When
Direct	Assume P, derive Q	P is strong/useful
Contrapositive	Assume $\neg Q$, derive $\neg P$	$\neg Q$ is easier to exploit
Contradiction	Assume $\neg S$, derive $C \wedge \neg C$	Statement involves impossibility
Cases	Split into $C_1 \cup \cdots \cup C_n$, prove each	Domain has natural partitions
Existence (C)	Exhibit witness x_0	"There exists" — can find one
Existence (NC)	Argue without finding x_0	"There exists" — can't find directly
Uniqueness	Existence + assume two, show equal	"Exactly one"
Vacuous	Hypothesis P is always false	Impossible precondition

Proof Writing Checklist

- ☐ State which method you're using at the start
- ☐ Clearly state all assumptions
- ☐ Justify every step
- ☐ Use definitions precisely (even $\Rightarrow$ write $n = 2k$)
- ☐ End with a clear conclusion statement
- ☐ Put ■ or Q.E.D. at the end
- ☐ For uniqueness: prove BOTH existence AND uniqueness
- ☐ For cases: make sure cases are exhaustive

EXAM PRACTICE

20 Questions — MCQ + Written ★ Good Luck Tomorrow! ★

PART A — Multiple Choice Questions (1–10)

Q1. Which of the following is logically equivalent to $P \Rightarrow Q$?
(a) $Q \Rightarrow P$ (b) $\neg P \Rightarrow \neg Q$ **(c)** $\neg Q \Rightarrow \neg P$ (d) $P \wedge \neg Q$

Q2. A proof that begins "Assume n is odd, then $3n + 5$ is even..." and derives "n is odd" from the conclusion side is using which method?
(a) Direct Proof **(b) Contrapositive** (c) Contradiction (d) Vacuous

Q3. To prove "$\sqrt{3}$ is irrational" using contradiction, the first line should be:
(a) Let $\sqrt{3} = p/q$ where p, q are both even
(b) Suppose $\sqrt{3} = p/q$ in lowest terms, with $p, q \in \mathbb{Z}$, $q \neq 0$
(c) Let $p = 2k + 1$
(d) Suppose 3 is rational

Q4. The statement "For all $x \in \mathbb{R}$, if $x^2 + 1 = 0$ then $x = 100$" is:
(a) False (b) True only for $x = 100$ **(c) Vacuously true** (d) Cannot be determined

Q5. A **constructive** existence proof for $\exists x \in \mathbb{Z},\ x^2 = x$ would:
(a) Assume no such x exists and derive a contradiction
(b) Exhibit a specific value of x and verify it
(c) Show $x^2 = x$ holds for all x
(d) Use the Intermediate Value Theorem

Q6. When proving "n^2 even $\Rightarrow$ n even", the best method is:
(a) Direct (b) Cases **(c) Contrapositive** (d) Vacuous

Q7. In a uniqueness proof of $\exists! x\ P(x)$, you assume $P(x_1)$ and $P(x_2)$ hold. The goal is to show:
(a) $x_1 = 0$ (b) $x_2 = 0$ **(c)** $x_1 = x_2$ (d) $P(x_1) \wedge P(x_2) = \text{False}$

Q8. Proof by cases requires that the cases be:
(a) Exactly two (b) Mutually exclusive only
(c) Exhaustive (cover all possibilities) (d) Each proved by contradiction

Q9. Which proof method would you use to prove: "There is no largest prime

number"?
(a) Direct (b) Contrapositive **(c) Contradiction** (d) Vacuous

Q10. The contrapositive of "If it rains, the ground is wet" is:
(a) If it doesn't rain, the ground is dry
(b) If the ground is wet, it rained
(c) If the ground is not wet, it didn't rain
(d) If it doesn't rain, the ground is wet

PART B — Written / Proof Questions (11–20)

Q11. [Direct Proof]
Prove that if m and n are both even integers, then $m + n$ is even.

Q12. [Direct Proof]
Prove that if n is odd, then n^2 is odd.

Q13. [Contrapositive]
Prove: If n^2 is odd, then n is odd.
(Hint: use the contrapositive)

Q14. [Contrapositive]
Prove: If ab is even, then a is even or b is even.
(Hint: contrapositive is "if a is odd AND b is odd, then ab is odd")

Q15. [Contradiction]
Prove that $\sqrt{3}$ is irrational.

Q16. [Contradiction]
Prove that there is no rational number r such that $r^2 = 2$.

Q17. [Proof by Cases]
Prove that $n^2 \geq n$ for all integers $n \geq 0$.
(Hint: consider $n = 0$, $n = 1$, $n \geq 2$)

Q18. [Existence Proof]
Prove that there exist integers m and n such that $3m + 7n = 1$.

Q19. [Uniqueness Proof]
Prove that the equation $5x - 3 = 12$ has a unique solution.

Q20. [Identify the Method — Short Answer]
For each statement below, state which proof method you would use and briefly justify:
(a) "If p is prime and $p > 2$, then p is odd."
(b) "There exists a real number x such that $x^3 = x$."
(c) "For all $x \in \mathbb{R}$, if $x^2 = -1$ then $x = 0$."
(d) "The integer k satisfying $2k = 10$ is unique."

Answer Hints (Written Questions)

- **Q11:** $m = 2a, n = 2b \Rightarrow m + n = 2(a + b)$
- **Q12:** $n = 2k + 1 \Rightarrow n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$
- **Q13:** Contrapositive: n even $\Rightarrow n^2$ even (direct)
- **Q14:** Contrapositive: $a = 2j + 1, b = 2k + 1 \Rightarrow ab = 4jk + 2j + 2k + 1 = \text{odd}$
- **Q15:** Assume $\sqrt{3} = p/q$ lowest terms, get $3 \mid p$ then $3 \mid q$, contradiction
- **Q16:** Same structure as $\sqrt{2}$ irrational proof
- **Q17:** Case $n = 0$: $0 \geq 0$; Case $n = 1$: $1 \geq 1$; Case $n \geq 2$: $n^2 \geq 2n > n$
- **Q18:** Constructive: $m = 5, n = -2$: $15 - 14 = 1$ ✓
- **Q19:** Existence: $x = 3$; Uniqueness: $5x_1 - 3 = 12 = 5x_2 - 3 \Rightarrow x_1 = x_2$
- **Q20:** (a) Contrapositive/Direct, (b) Constructive Existence ($x = 0, 1, -1$), (c) Vacuous, (d) Uniqueness

You've got this! — Good luck on your exam tomorrow ★

Remember: state your method · justify every step · end with ■